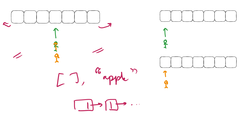
# Babua DSA Patterns Course

## Day 1/90

Premium

Comment "Day 1 - Done"

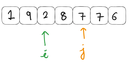
### Two Pointer Pattern



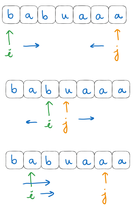

### Two Pointer Pattern

→ A technique where two variables (pointers) traverse a data structure (like an array, string, or linked list) simultaneously from different positions

Linear Data structures




→ Two pointers either move towards each other, away from each other, or in the same direction

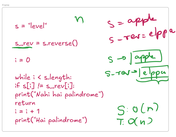



→ Efficiently solving problems involving comparisons, searching or partitioning

PCS

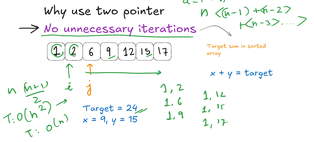
Palindrome

palindrome? ✓



Why use two pointer

→ No unnecessary iterations




→ Early stopping as soon as mismatch/condition fails

→ Works in-place - no need to create new arrays or copies

→ Improves both speed and space efficiency

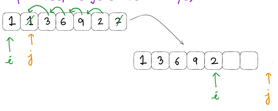
When to use it?

→ Comparing elements from both ends (palindrome, pair sum)

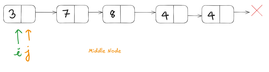
PCS

→ Finding pairs with specific properties (e.g, sum = target)

→ Removing or merging elements (e.g., remove duplicates, merge sorted arrays)




→ Detecting loops or middle elements (e.g., linked list cycle, middle node)




Template

### 1 Input, Opposite

```
a = [1, 4, 3, 2, 9]
left = 0
right = a.length - 1

while left < right:
if CONDITION:
left = left + 1
else:
right = right - 1

return ans
```

### 2 Inputs, Exhaust both

```
a = [1, 4, 3, 2, 9]
b = [9, 7, 6, 2, 0, 8]

left = 0
right = 0

while left < a.length and right < b.length:
if CONDITION:
left = left + 1
else:
right = right + 1

while left < a.length:
left = left + 1

while right < b.length:
right = right + 1
```

1 4 6 9 10
9 18 16 22 30
[1 4 6 9 1
10 10 16 22 30]

DSA In 90 days